Internship Project 2: Static Structural Analysis of Conventional Sugar Mill Roller Shaft

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Static Structural Analysis of Conventional Sugar Mill Roller Shaft

1. Abstract

In the course of my Mechanical Engineering internship, I undertook the static structural analysis of the sugar mill roller shaft through ANSYS. In the sugar industry, crushing sugarcane to extract juice is much dependent upon the roller shaft, which most of the time suffers heavily under compressive and torsional forces. Therefore, there is a need for the shaft to be scrutinized for mechanical strength and reliability.

The project comprised creating a 3D model of the shaft in ANSYS Design Modeler, assigning material properties, imposing realistic boundary conditions, and running static structural simulations. The objectives were to grasp the behaviour of the shaft when subjected to applied load, mainly concerning stress distribution, deformation, and safety of load.

The report details each step of the entire process-from generic concept formation, modelling, and simulation-to interpretation of results. This pronounced my FEA skills and deepened my understanding of real-world engineering problem-solving through simulation software.

2. Introduction to Sugar Mill Roller Shaft

I studied the mechanical components used in sugar milling operations during my internship. Among such elements, the sugar mill roller shaft grabbed my attention due to its size, complexity, and operational significance.

2.1 Purpose of the Shaft in Sugar Industry

The shaft connects to the rollers that crush sugarcane. Torque is transmitted from a motor or gearbox to initiate rotation. Sugarcane is fed into the rollers, which compress it with almost huge force to juice it out.

2.2 Real-World Considerations

In-the-real-world considerations for the shaft require that it should:

- Bear the full load of the roller and cane pressure.
- Transmit torque without twisting or breaking.
- Operating continuously with almost no maintenance.

From my field inspection, I learned that shaft failure is most commonly a consequence of fatigue, keyway cracking, or bearing misalignment. Hence, simulation of shafts under load conditions is used to enable the engineer to anticipate any trouble and to redesign for an improvement in durability.

3. Brief about Ansys Software

Throughout the internship, I was familiarized with ANSYS Workbench, an advanced engineering simulation software employed by industries for computer-based verification of mechanical systems.

3.1 My Experience with ANSYS

First, I was taught the interface and modelling environments, followed by solving engineering problems in the real world. ANSYS enabled me to carry out:

- Geometry modelling of mechanical parts.
- Mesh generation with element size and shape control.
- Mesh generation with stress and deformation simulation.
- Post-processing to visualize the crucial regions and extract numerical values.

3.2 Industry Significance

ANSYS is applied by engineers to cut down prototype testing, enhance safety, and maximize design. For instance, any small modification in fillet radius or shaft diameter can be rapidly simulated, conserving time and expenditure in actual testing.

This project made me realize how classroom theory (such as stress-strain analysis) can be transferred in real engineering practice utilizing simulation software.

4. Modelling of the sugar mill roller in design Modeler

This topic describes how I developed the 3D CAD model of the roller shaft in ANSYS Design Modeler.

4.1 Planning of Geometry

I modelled the approximate dimensions of the shaft from typical sugar mill drawings and dimensions given by my industry mentor prior to modelling.

4.2 Step-by-Step Modelling Process

Launch ANSYS Design Modeler

I launched the Design Modeler from Workbench under the "Geometry" module.

Set Units

Converted unit system to millimetres to align with industrial standards.

Sketch Shaft Profile

On the XY plane, I drew a 2D profile in terms of lines and arcs.

Added several stepped diameters to indicate bearing areas, roller mounting areas, and coupling ends.

Create 3D Model Using Revolve

Revolved the 2D sketch by 360° around the central axis to create a solid shaft.

Add Keyway Feature

Swept in a rectangle at the end of the shaft to depict the keyway.

Applied Boolean subtraction to eliminate this slot from the solid body.

Apply Fillets

Added fillets on diameter changes to prevent sharp corners that might lead to stress concentrations.

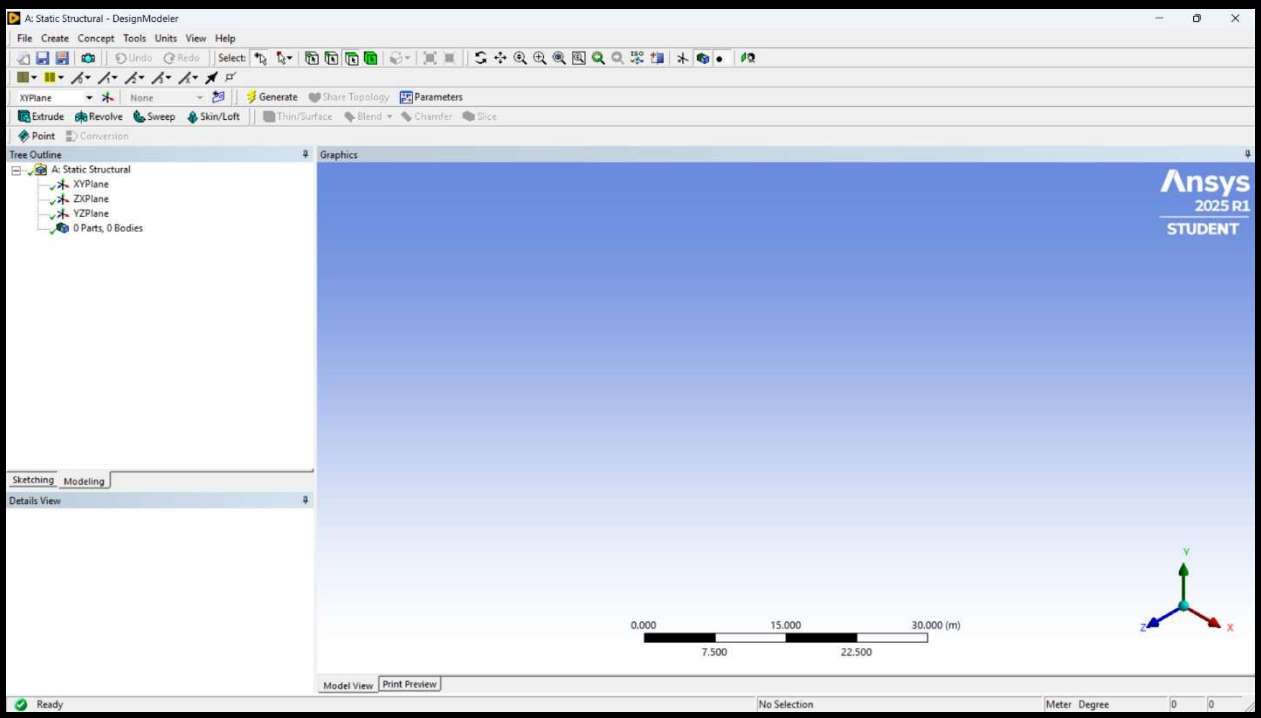
Inspect and Finish Model

Rotated and zoomed out the model in order to check for geometric accuracy and symmetry.

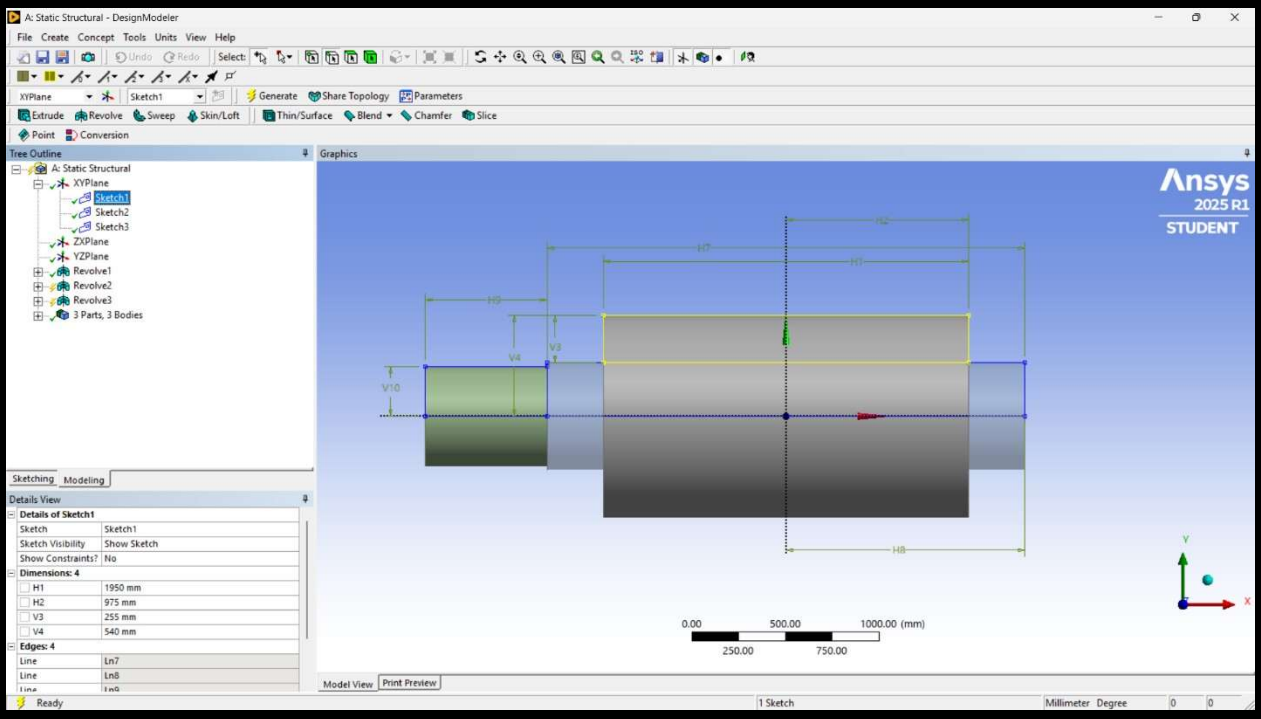
Save and Export saved geometry and updated it within the Workbench project.

4.3 Learning Outcome

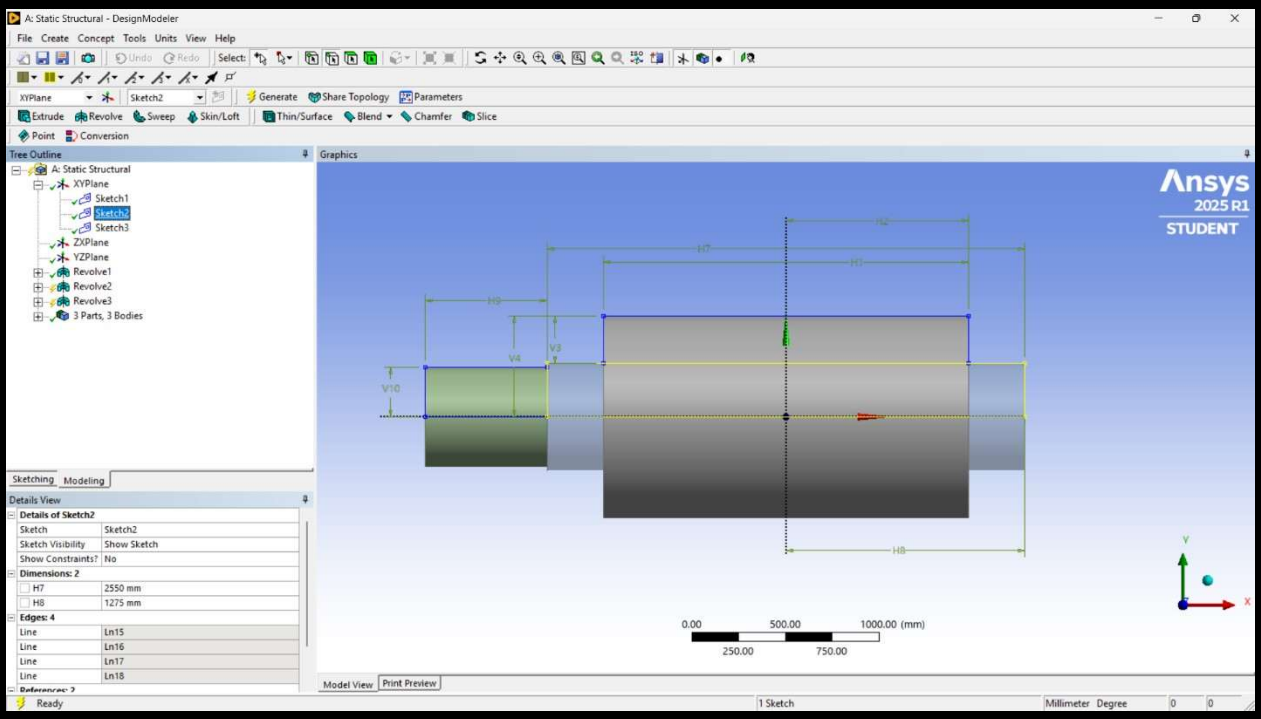
This hands-on learning experience instilled in me the significance of precision modeling. A small error in geometry (such as a missing fillet or extra edge) would produce improper meshing or faulty simulation results.



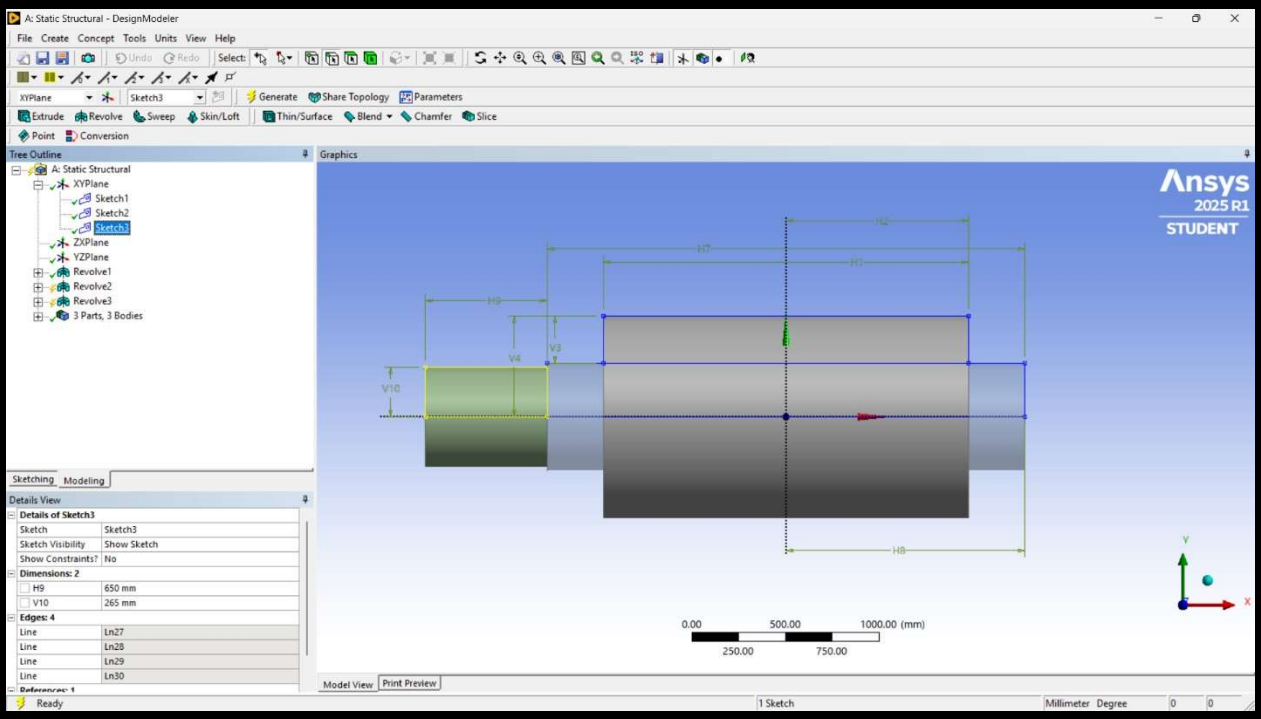
This typically is the opening interface of Design Modeler from geometry in Static Structural of Ansys Workbench. Options like sketching and modelling are present, which provides us with various sketching tools and other options. The workspace where the model will be present is of a normal cartesian system consisting 3 principal axes.



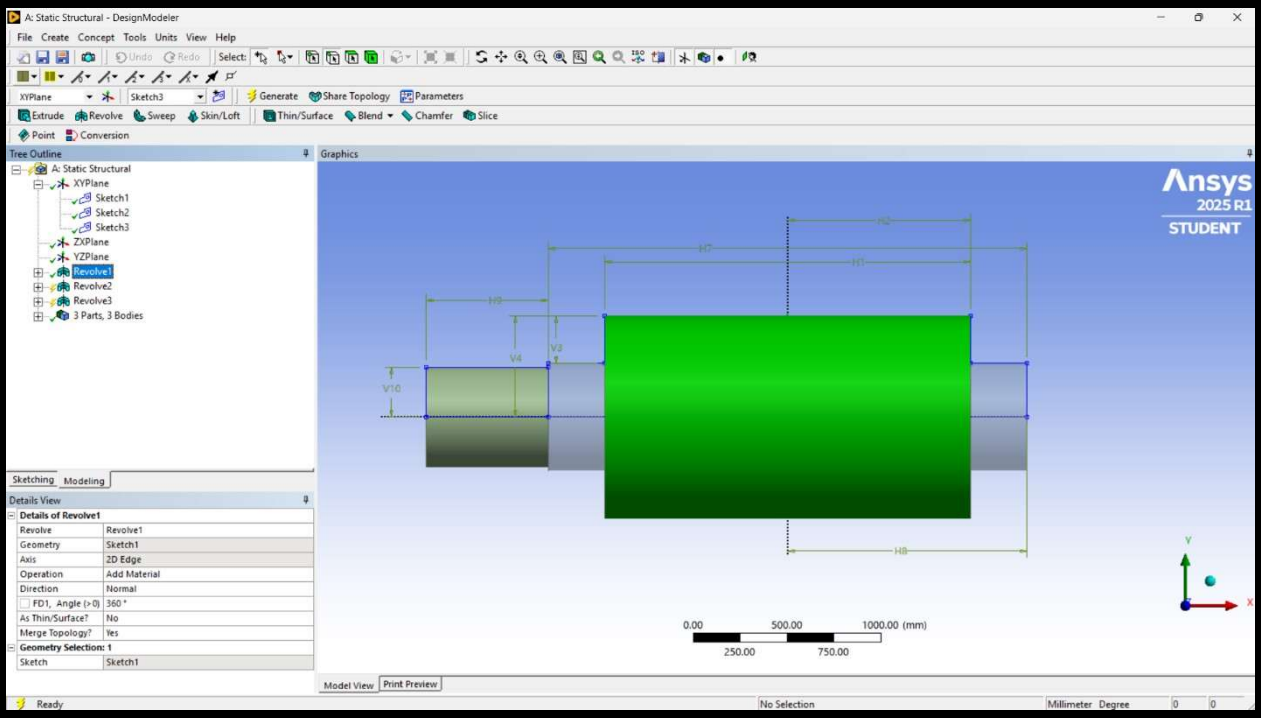
Now, the image given shows the 3D model of the Shaft which is almost completed but the highlighted yellow rectangle is the part the image is actually meant for. The rectangle is first drawn from the sketching, draw option, with proper dimensions for a convention sugar mill roller shaft. After that rectangle is drawn, we revolve the area with axis as the x axis, which will provide a cylinder with a hole.



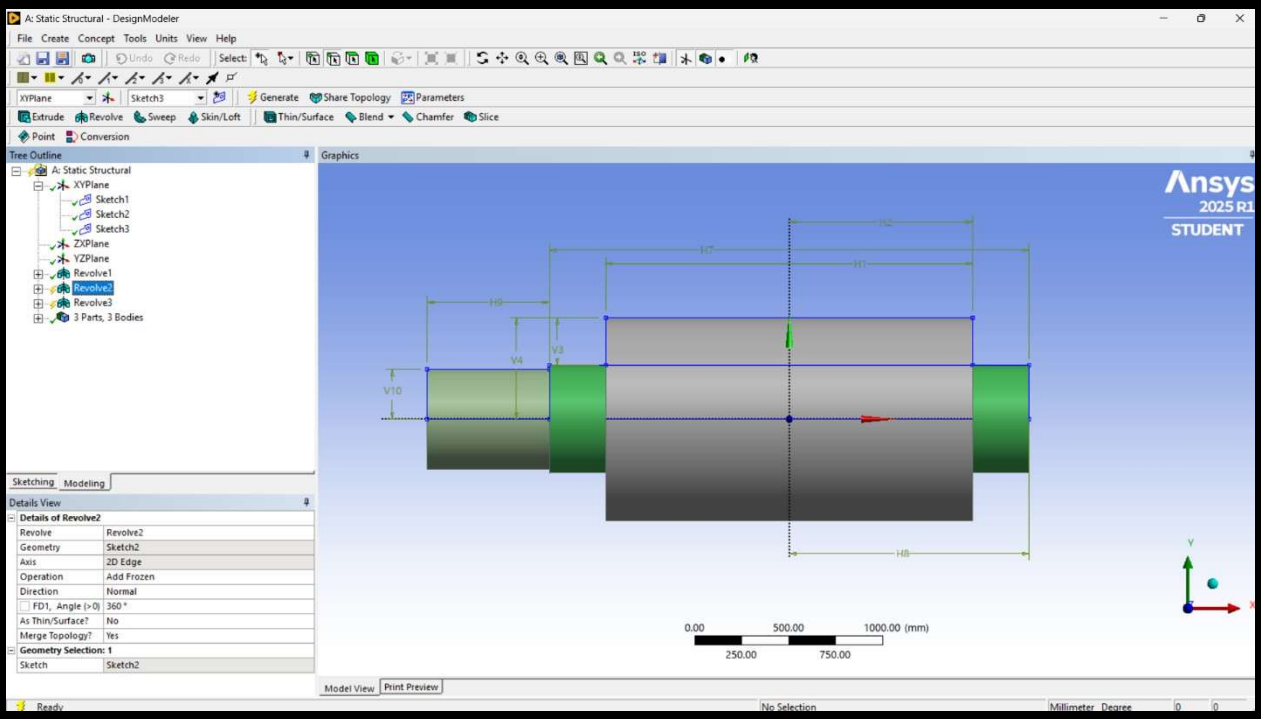
This image highlights the second rectangle which will form the next volume of the cylindrical shaft. It is done in the same way we did the first one but the only thing we need to take care is while revolving, we have to change from add material to add frost, which will help us separate the two materials from each other, which in turn helps us differentiate the material used in their preparation.



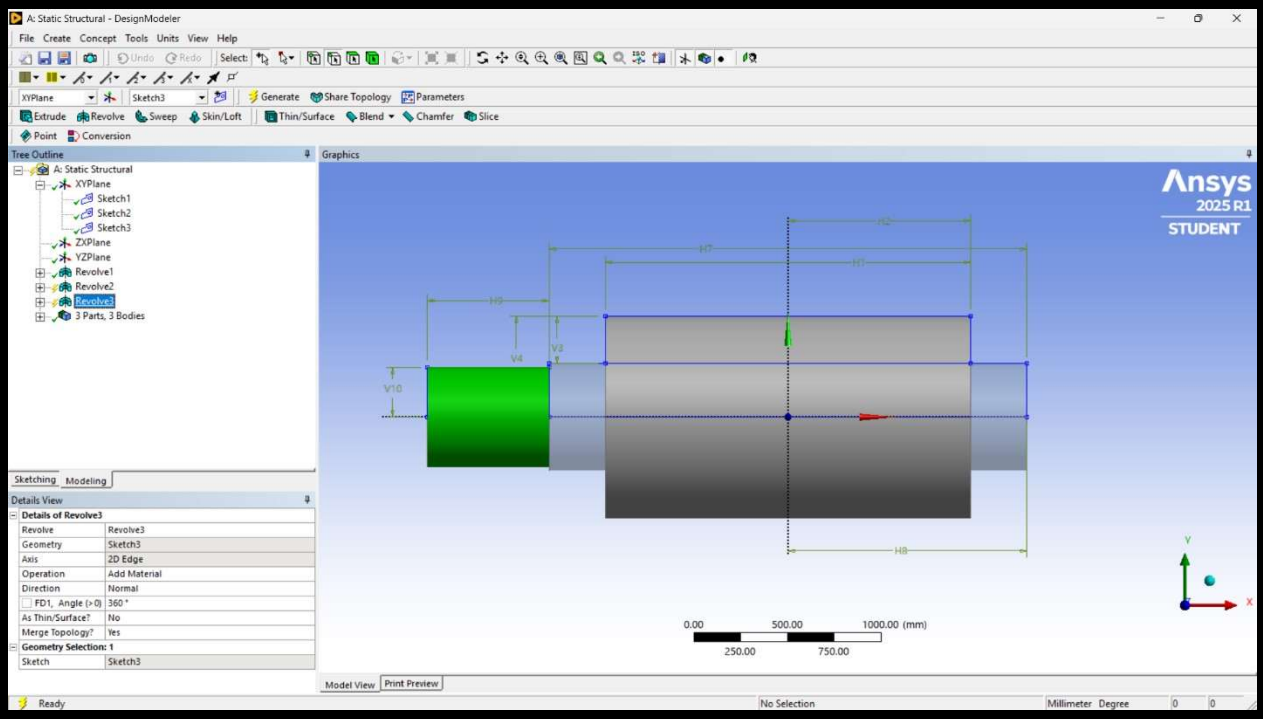
After sketch 2, change it to the next sketch and draw another rectangle on the left side with conventional dimensions. Revolve it again to get the next part of the shaft.



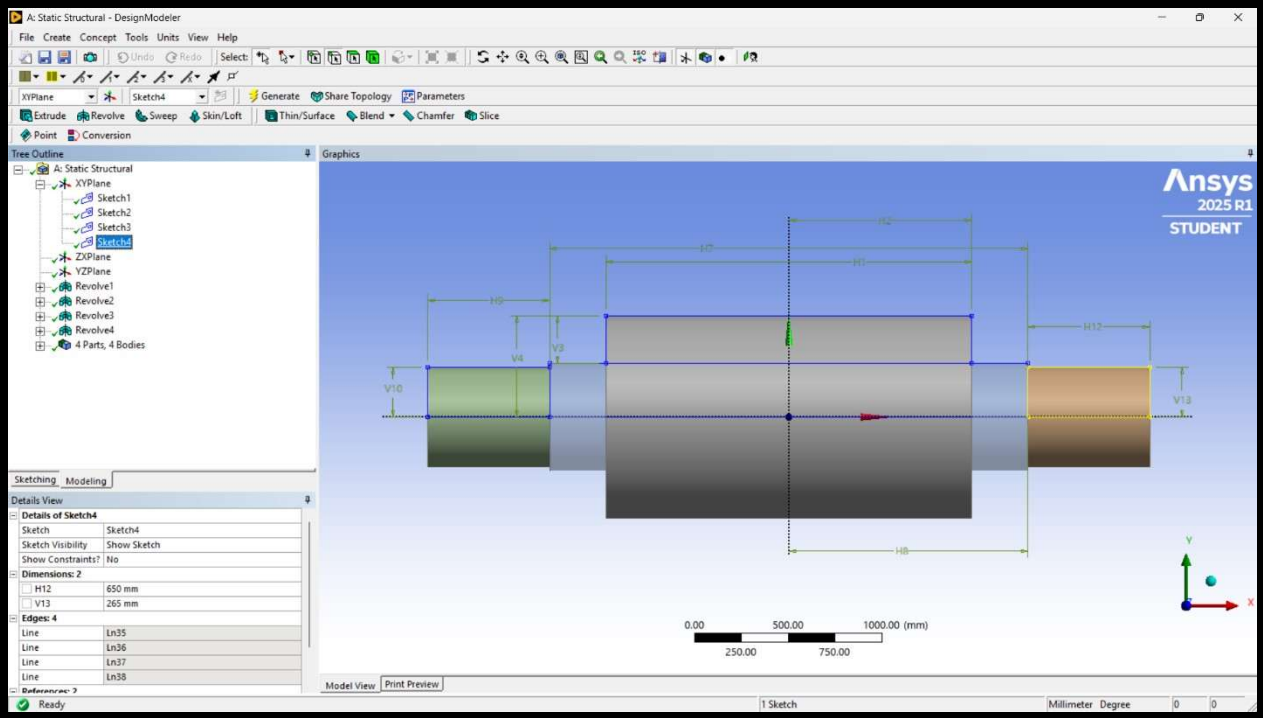
From the first rectangle, I revolved the area and thus the green volume is created. While creating, select the rectangle, then opt revolve and then select the axis by which we need to revolve it, here its x axis for me. Moving it in different directions, we will be able to find the formation of the shaft kind of holed cylinder.



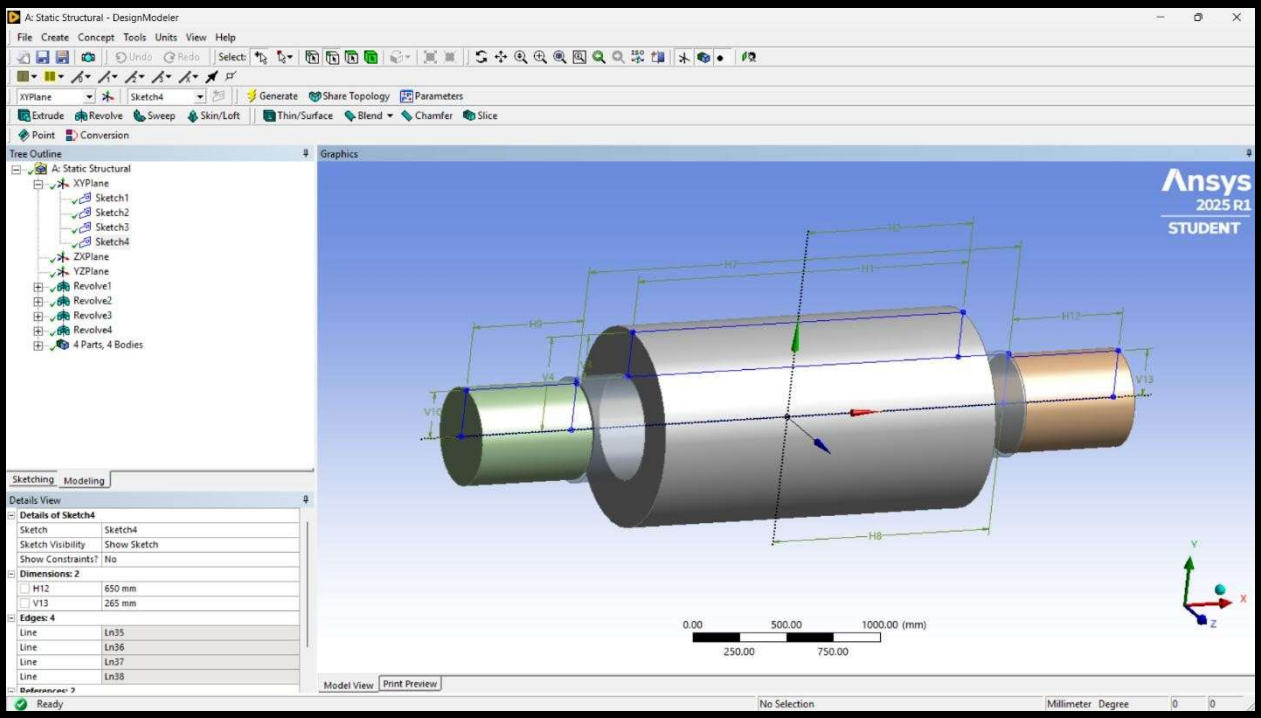
This image suggests the green coloured shaded region as a result of revolve 2. This will now remove the whole and help in forming the shaft to a higher extend.



The rectangle to the left side of the shaft formed is being revolved to produce the green shaded region as shown in the image.



Same procedure is used again on the right side, which is shown as the brown coloured cylinder which will complete the modelling of our conventional Sugar Mill Roller Shaft. The next image will provide a better view for proper understanding.



The image provides with a proper view for the shaft with different colours designated to different parts of the sugar mill roller shaft. All the dimensions are clearly visible till now and the geometry part of structural analysis of the shaft has been successfully done using the Design Modeler. Various features from sketch 1 to 4 and simultaneous revolved parts from 1 to 4 has been shown in the right column, bottom to which gives us the information that the shaft is made of 4 parts, 4 bodies. This way of modelling with different parts allows us use different materials at different sites and hence improving the texture, strength of the shaft and make manufacturing easier. As the Geometry of the Conventional Sugar Mill Roller Shaft has been done, we can now go to the next step of model and rest of them.

5. Structural analysis of sugar mill roller shaft using Ansys

Following modelling, I proceeded to static structural analysis using ANSYS Mechanical.

Import Geometry

Connected the previously created shaft into the "Static Structural" system.

Material Assignment

Selected "Structural Steel" from the material library:

Young's Modulus = 200 GPa

Poisson's Ratio = 0.3

Yield Strength = 250 MPa

Apply Boundary Conditions

Fixed both ends to mimic bearing constraints.

Applied axial force on roller region to mimic sugarcane pressure.

Optionally added torque to mimic real-time shaft operation.

Generate Mesh

Applied default mesh with refinements in keyway and fillets.

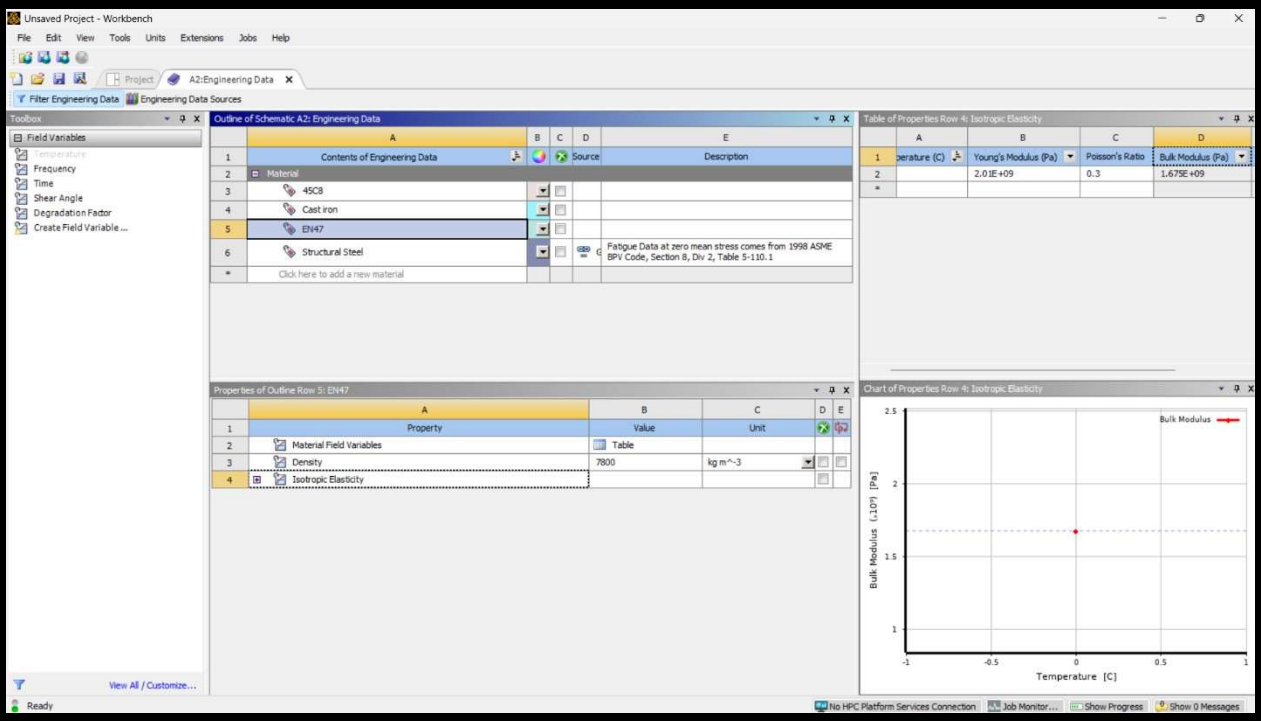
Checked element quality to ensure precision.

Solve

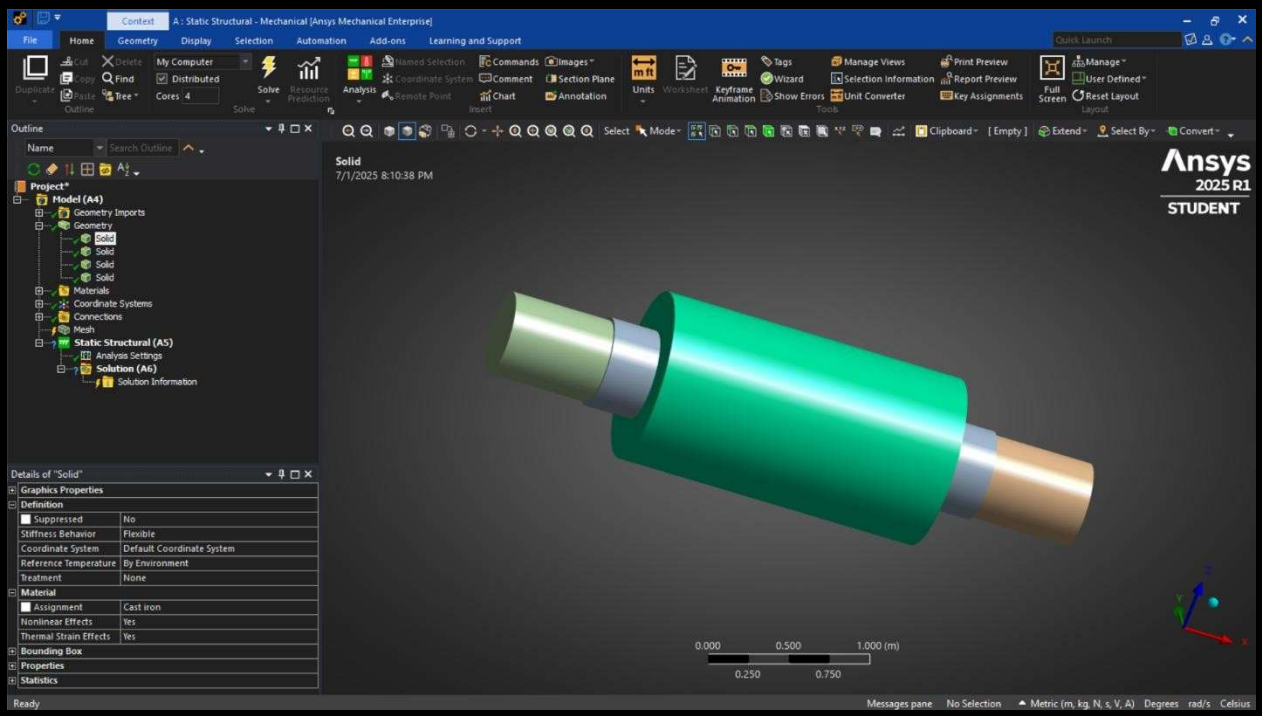
Solved the model. Solver runs without errors.

Viewed real-time deformation and stress contours during post-processing.

And with these steps we will be able to run the analysis of the shaft and hence select which material can be more apt for places like this.

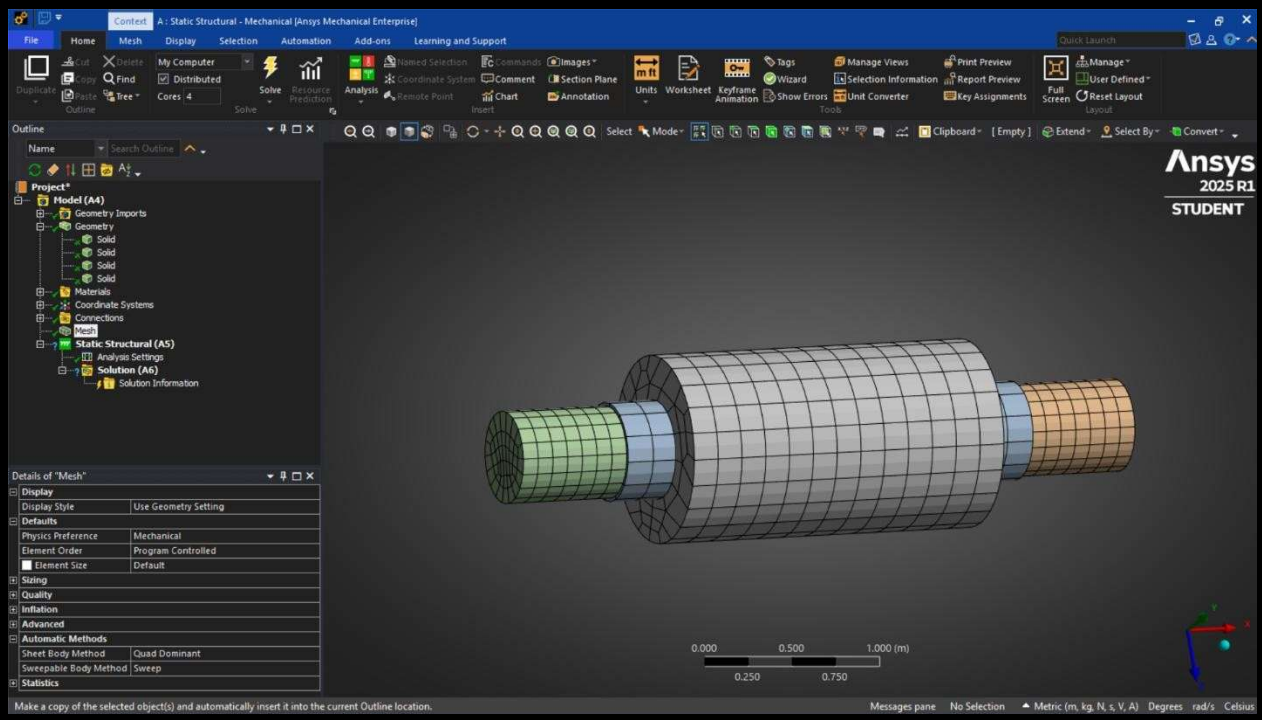


After completing the geometry, we went to add further more materials in the engineering data section. Here we added Cast iron used at the centre portion of the shaft in every case, 45C8 and EN47 at the end parts. The materials, whose properties like Density, Young's Modulus and Poisson's ratio has been added from other resources. With the use of the given details, the software itself calculates the rest of the properties and thus help us know the strength and other features which in turn helps us to select what material can work on which site and all.

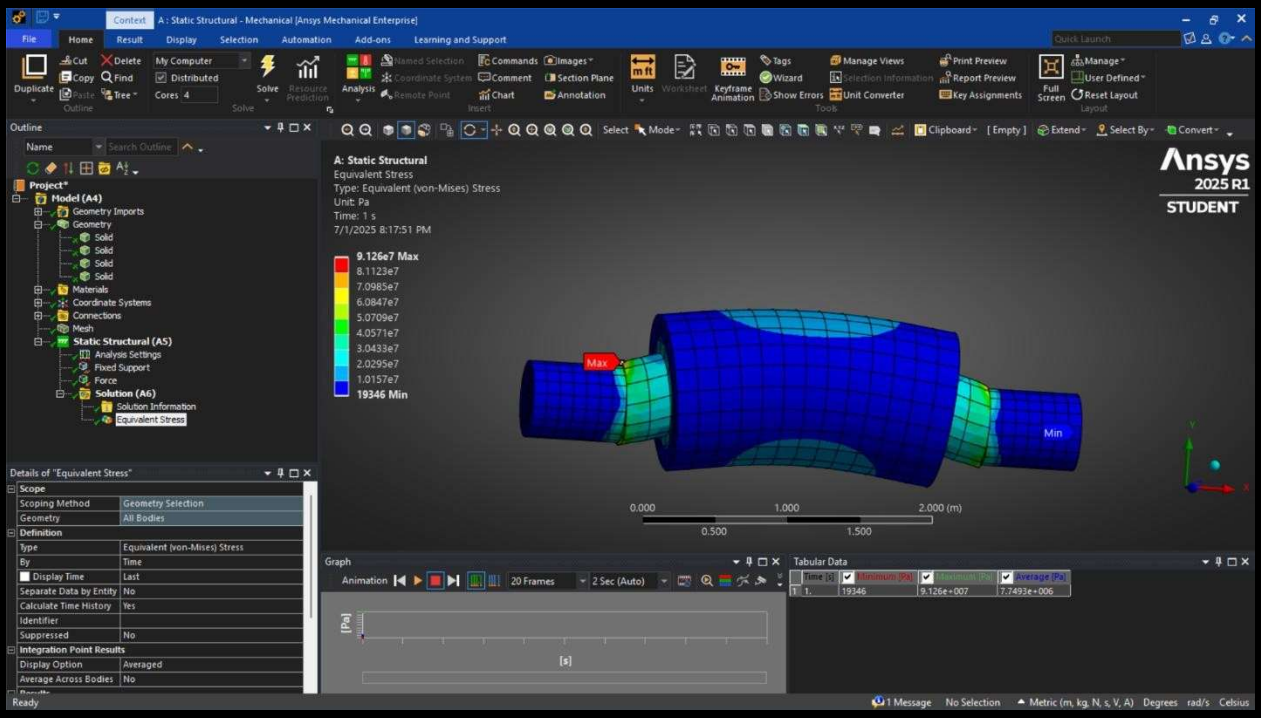


The image shows the shaded region which is made of Cast iron as said earlier.

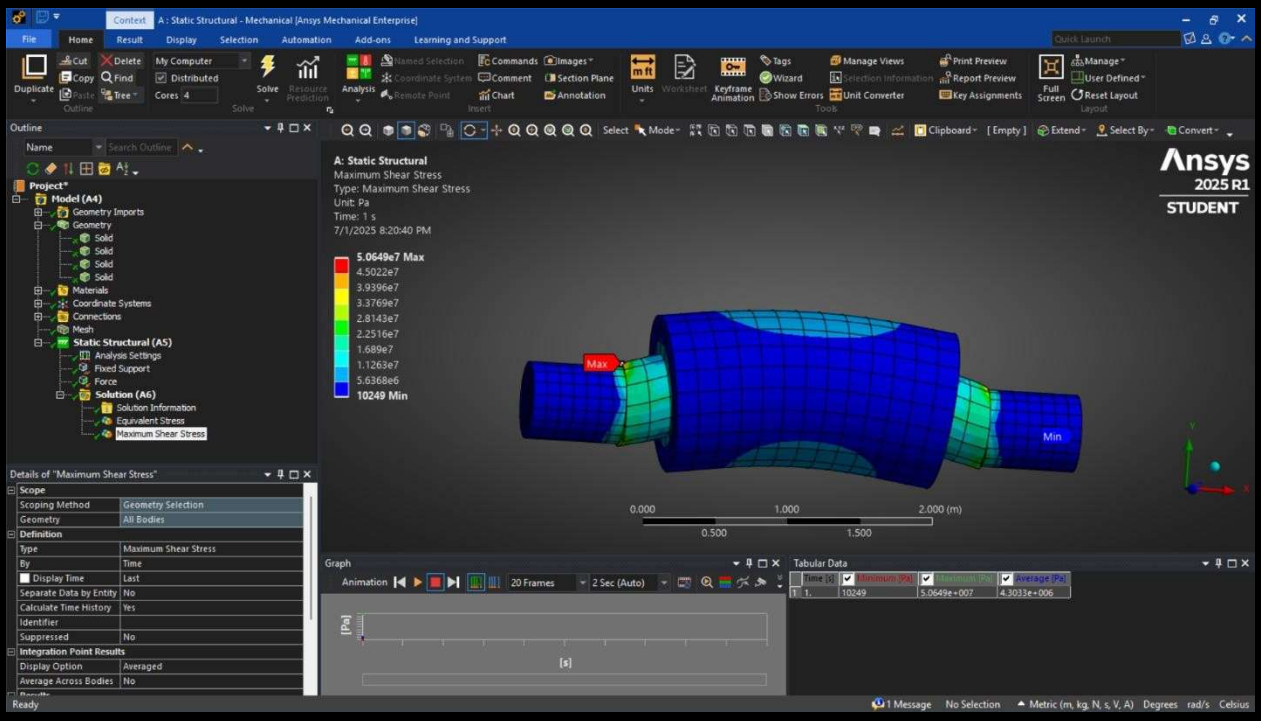
In the first case, the rest of the parts are made of 45C8.



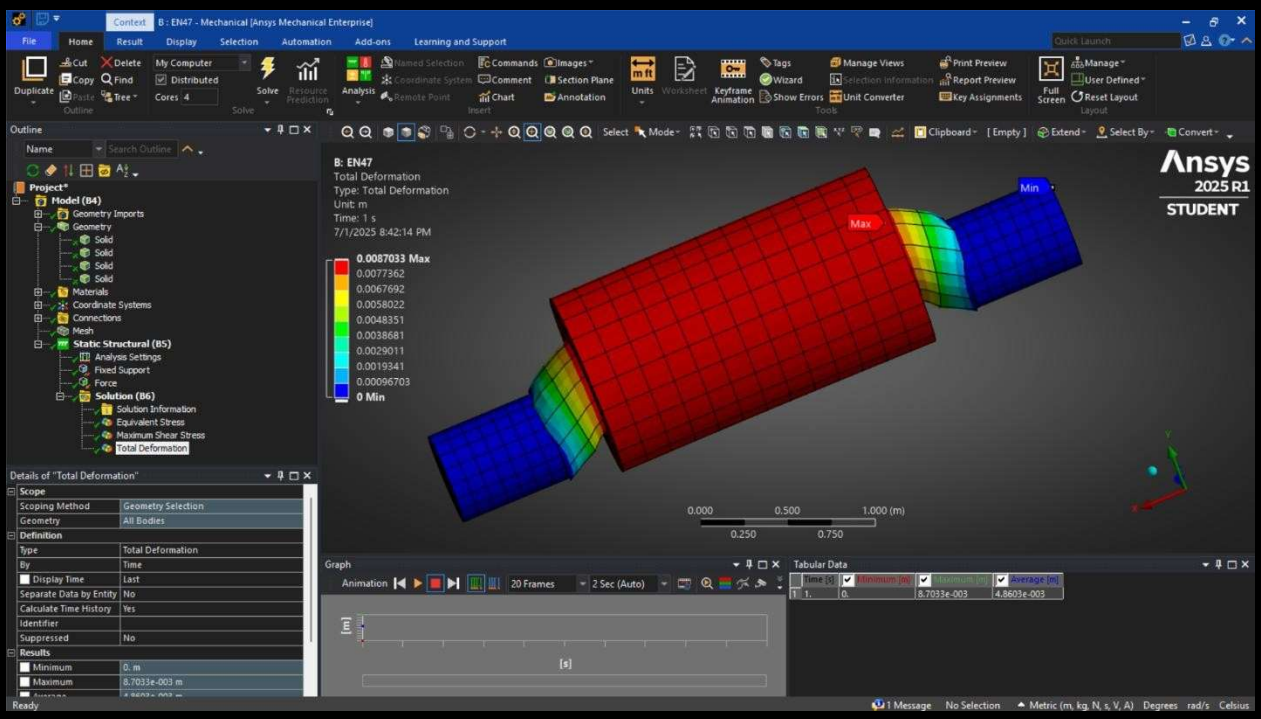
An expanded mesh has been generated for the shaft, which breaks down each single component into various unit components. The difference in the pattern is related with the difference in features and properties like diameter and all. Now at the left side above the details tab, opt for static structural from where, opt insert option which will lead to a number of options. From there primarily fix the both ends of the shaft. Choose fixed ends and select the side most cylindrical part of the shaft, apply the surface and thus they will get selected. Moving further, right click on static structural again and from insert select force. Now select the surface where we apply the force and give the value for the force which we r taking as 760 tons equivalent to 7572652 N. Now go for solution and the software will do the required calculations, which will take a bit of your time. Once u r ready with the results, start analysing with each other.



The software has calculated the results and the image shows the animation of what will likely happen in the application of a certain force. Here the force resulted in an maximum equivalent stress of 9.126E7Pa.

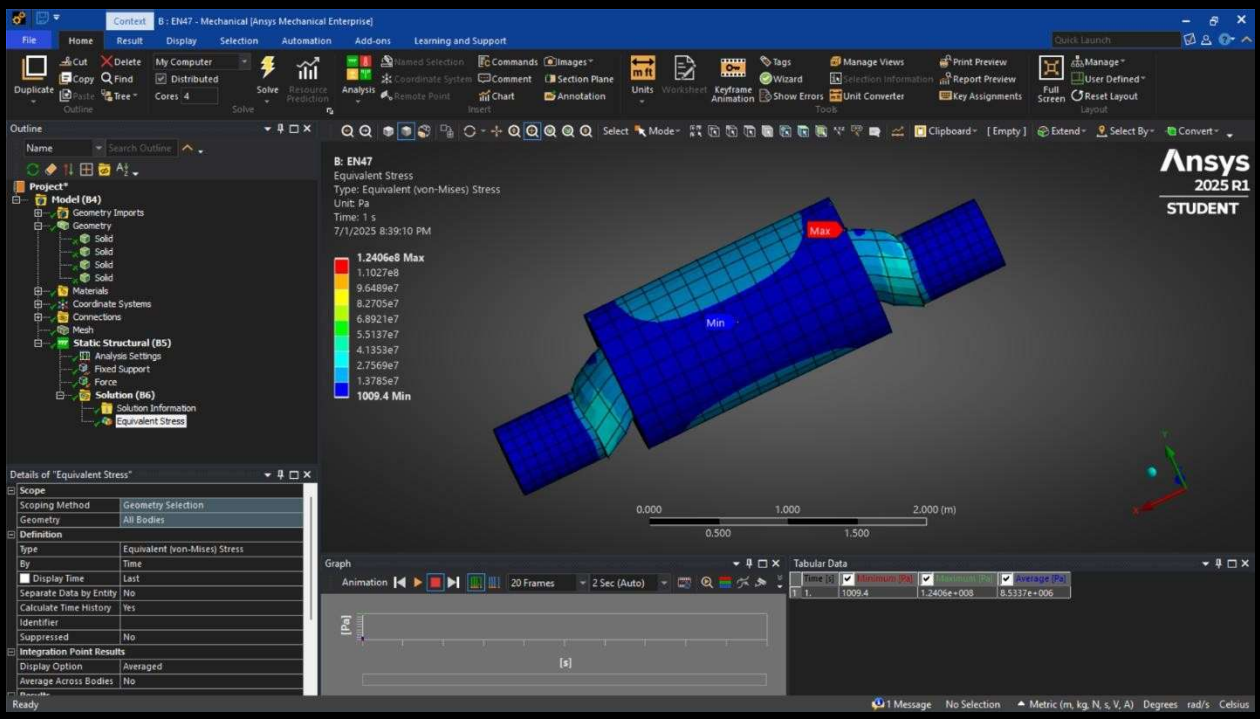


The same force created a maximum shear stress of 5.0649E7Pa for the same material of 45C8.

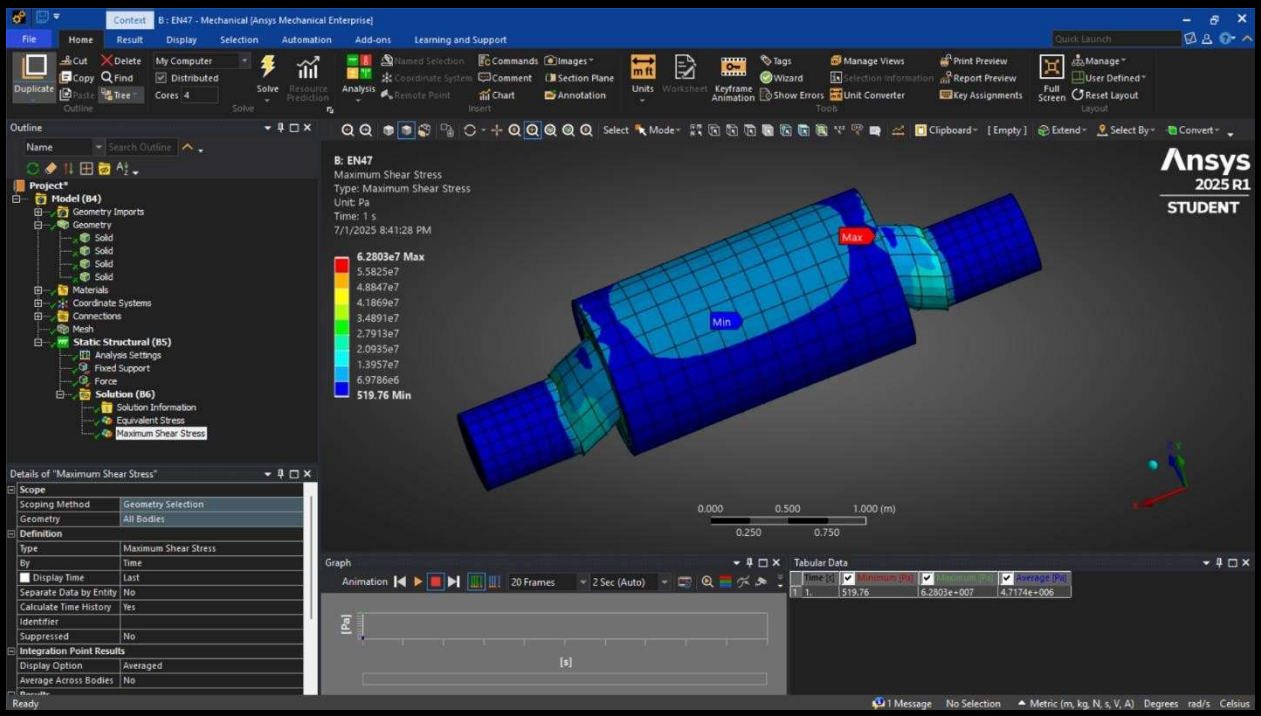


From the image its also clear that the same force on 45C8 material created a deformation of o.oo87m, which considerably is a small value.

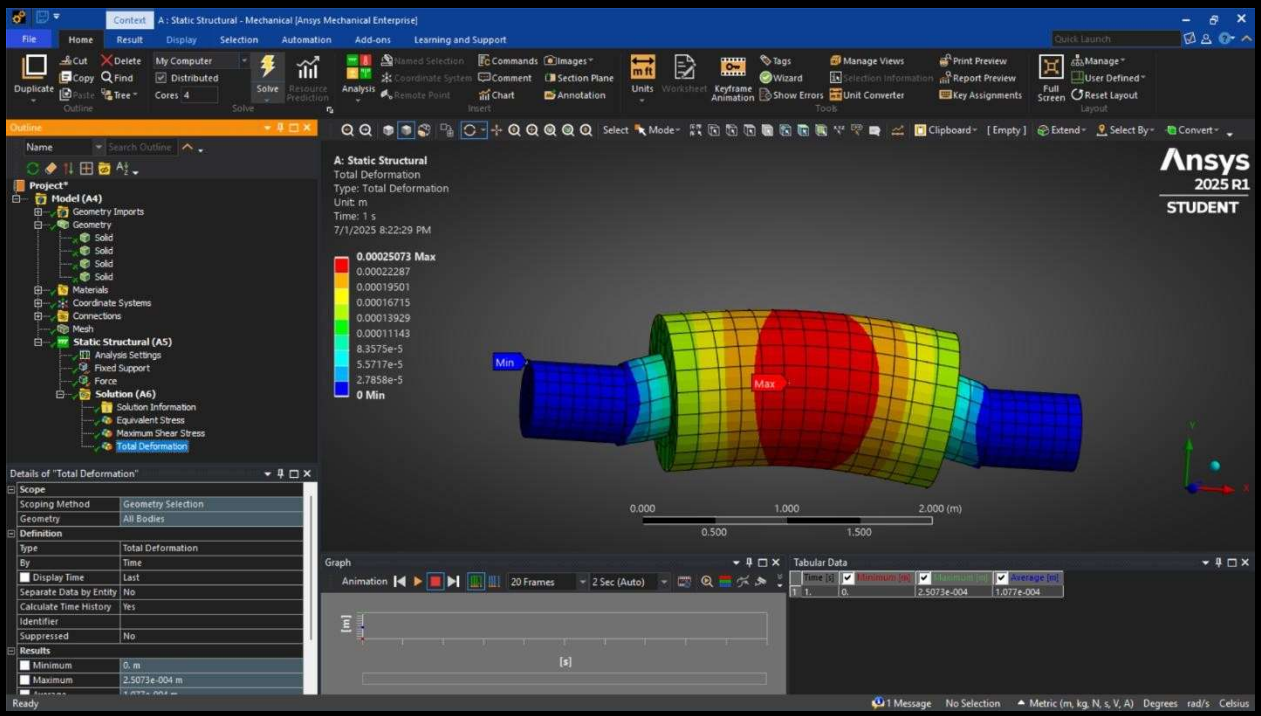
In the second case, the rest of the parts are made of EN47.



A higher equivalent stress of 1.2406E8 Pa has been seen in material made of EN47.

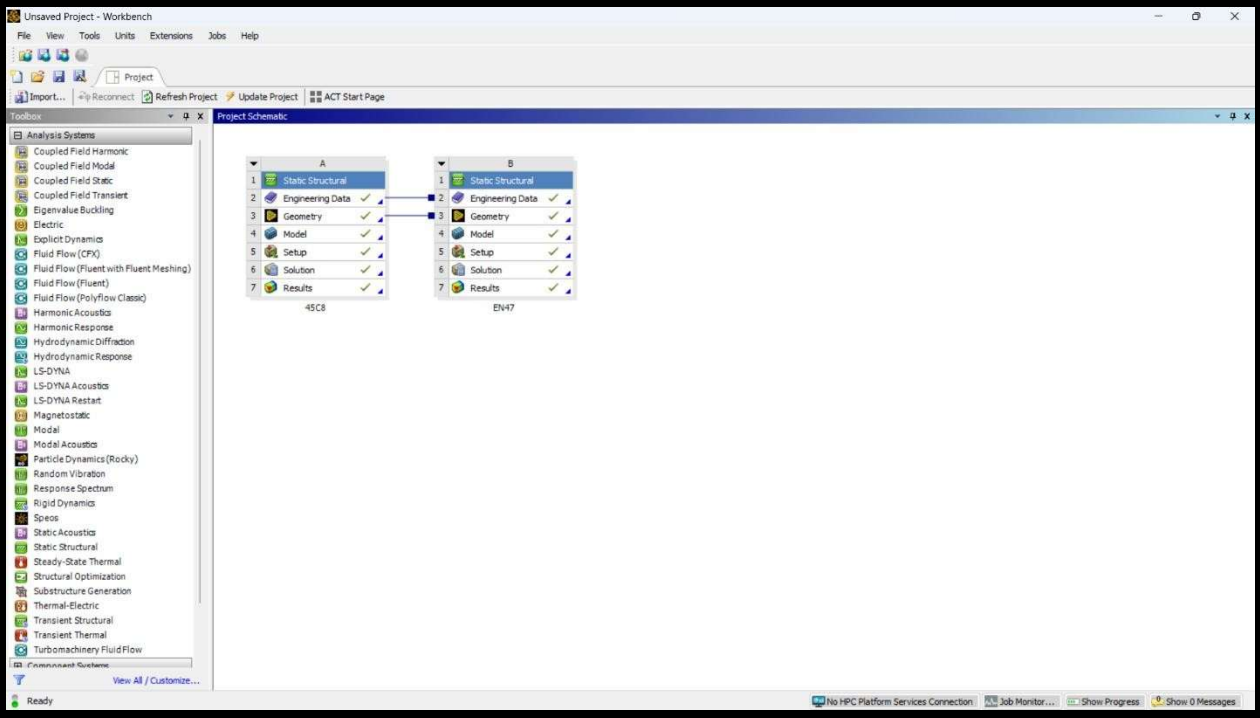


A higher maximum shear stress has been seen.



A lower value of deformation has been seen again for EN47.

All this specifies that EN47 is a stronger and better material for the shaft.



and thus, the analysis comes to an end.

6. Results

Parameter Value Units

Max Equivalent Stress (von-Mises) 216 MPa MPa

Material Yield Strength 250 MPa MPa

Maximum Total Deformation 0.28 mm

Factor of Safety ~1.15 - 6.2

Results Interpretation

Stress Distribution: Maximum stress is seen at keyway and fillet areas—suggesting likely fatigue zones.

Deformation: Least amount of shaft bending, indicating good stiffness under loads applied.

Safety: The shaft is working below the yield strength, which implies it's structurally safe for the provided loading condition.

6.3 Personal Takeaway

This report emphasized the significance of concentration of stress, particularly about sharp corners and geometric changes. I realized how small modifications in design (such as fillet radius increase) can lower peak stress and improve shaft life.